Networks: getting the data home



IoT · Level 3 · Lesson 4

Today we will

- Compare Bluetooth, Wi-Fi, mobile networks and LoRa by range, power, data and cost.
- Choose a network for a school sensor from its distance, power and data needs.
- Explain how gateways and relays (a mesh) carry messages beyond one radio's reach.

Choosing a network

- Ask: how far, what power, how much data, what cost?
- Near: Bluetooth LE or Wi-Fi
- Far: a mobile network or LoRa
- Out of reach: a gateway and relays

The playground radio game

1. Warm-up (5 min): The tank sensor nobody can hear.
2. Teach (8 min): Four questions and four radios.
3. Play (15 min): Pass notes hop by hop to the gateway.
4. Choose (8 min): Draw the school; choose a network for one sensor.
5. Wrap-up (4 min): What happens when a relay fails?

Choose a safe, open space and agree a signal to stop. Pupils walk, never run, and pass notes only. A pupil who can't move about easily can be the gateway.

Let’s discuss

1. Which school sensor would you connect first, and how?
2. What happens to the data if the gateway loses power?
3. When would a memory card collected weekly be enough?

Exit ticket

- A soil sensor 2 km from school, on batteries, sends 10 bytes an hour. Which network, and why?